

Lab 4: LLM Usage Report

Memory Color Pattern Repair Lab

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LLM used	Claude (Anthropic), chat interface
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Project	Pygame Simon-style game: fix 1 bug, implement 3 features

1. What I used the LLM for

I used the LLM as a pair-programming and debugging partner, as the lab asks. I gave it the lab README and my `main.py`, and it produced the code for all four tasks. It also tested the code, helped me work out why the game would not start on my machine, and helped me prepare the GitHub submission.

Task	What the LLM did	How it was checked
1. Fix sequence bug	Rewrote round advance so each round appends exactly one random color to the sequence.	Automated run over 4 rounds: sequence length was 1, 2, 3, 4 and score 4.
2. Playback acceleration	Flash time 600 ms down by 40 ms per round (min 150 ms). Pause 300 ms down by 20 ms per round (min 80 ms).	Printed values: round 1 = 600/300 ms, round 10 = 240/120 ms.
3. Sound	Generated a sine tone per pad (C4, E4, G4, C5) in code, played on flash and on click, plus a low buzz on game over.	All four sounds created without error. Falls back to silent if no audio device.
4. Turn timer	Countdown bar (green, yellow, red). Limit = 3 s + 1 s per step. Time-out triggers Game Over.	Forced a time-out in a test: state changed to GAME_OVER.
Extra checks	Wrong pad gives Game Over. R resets to round 1, score 0.	Both confirmed in the automated test.

2. Conversation log

Summary of each exchange in order.

#	My message	What the LLM did
1	Uploaded the lab README and <code>main.py</code> , asked to change the code according to the README requirements.	Noted that <code>game_engine.py</code> and <code>color_button.py</code> were not provided, so it wrote full versions of <code>main.py</code> , <code>game/game_engine.py</code> and <code>game/color_button.py</code> covering Tasks 1 to 4. Ran headless tests (see section 1).
2	"now what shud i run"	Gave the commands (<code>pip install pygame</code> , <code>python main.py</code>), the expected folder layout, and what to show in the demo video.
3	"its ending without the game even starting"	Ran <code>main.py</code> itself for 5 seconds to check; it stayed open, so the code was not crashing. Listed likely causes on my side: wrong <code>main.py</code> , wrong folder, a traceback in the terminal, or only some files replaced.
4	Sent a screenshot showing "Memory Pattern Arena" and "WRONG PATTERN! GAME OVER".	Spotted that this text is not in the new code, so my old <code>game_engine.py</code> and <code>color_button.py</code> were still being used with the new <code>main.py</code> . Told me to replace both files.
5	Sent a VS Code screenshot of the Explorer.	Saw that the new files were in the project root instead of the <code>game/</code> folder. Told me to move them into <code>game/</code> (replacing the old ones), add an empty <code>__init__.py</code> , and remove a stray breakpoint.
6	"now how to push this my git"	Gave git status, add, commit and push steps, with fixes for common errors (no upstream, no remote, rejected push, login).

#	My message	What the LLM did
7	Sent a screenshot of my GitHub lab repo and said the game must be pushed there as a separate submission.	Explained that my game folder was linked to a different remote, so the files should be copied into a clone of the lab repo and pushed from there. Gave PowerShell commands.
8	Asked to push as a separate folder for Lab 4 so my teacher is not confused.	Gave commands to place everything in a Lab4_Memory_Color_Pattern folder, with the expected repo layout and a check with git status.
9	Asked for README content, said I have both video clips.	Wrote a README covering the bug fix, all four tasks, how to run, folder structure, expected behavior and the video and chat links.
10	"how to paste a video in readme"	Explained three options: drag and drop in GitHub's web editor (inline player), push to a videos/ folder and link, or link to Drive/YouTube.
11	Asked for a PDF of what I did with the LLM.	Generated this report.

3. Problems I ran into and how they were resolved

Problem	Cause	Fix
Game showed Game Over immediately with different text and title.	New main.py was running with the old game_engine.py and color_button.py.	Replaced both files in the game/ folder with the new versions.
Imports could not find the new files.	New files had been saved in the project root, not inside game/.	Moved them into game/ and added __init__.py.
Git remote pointed to a different repo than the lab repo.	The project folder was linked to another remote.	Cloned the lab repo and copied the project into a Lab4 folder there.

4. Key code written with the LLM

Task 1: one new step per round

```
def _start_round(self):
    # exactly ONE new step per round
    self.sequence.append(random.randrange(len(self.buttons)))
    self.state = WATCH
    self.input_index = 0
    self.play_index = 0
    self.phase = "intro"
    self.phase_start = pygame.time.get_ticks()
```

Task 2: playback speeds up with the round number

```
def flash_ms(self):
    return max(FLASH_MIN, FLASH_START - (self.round_no - 1) * FLASH_STEP)

def pause_ms(self):
    return max(PAUSE_MIN, PAUSE_START - (self.round_no - 1) * PAUSE_STEP)
```

Task 3: one tone per pad (sine wave, no external files)

```
def make_tone(freq, ms=300, volume=0.4, rate=44100):
    n = int(rate * ms / 1000)
    buf = array("h")
    for i in range(n):
        env = min(1.0, i / (rate*0.01), (n - i) / (rate*0.05)) # fade, no clicks
        s = int(32767 * volume * env * math.sin(2*math.pi*freq*i/rate))
        for _ in range(channels):
            buf.append(s)
    return pygame.mixer.Sound(buffer=buf.tobytes())
```

Task 4: turn timer

```
def turn_limit_ms(self):
    return TURN_BASE_MS + TURN_PER_STEP_MS * len(self.sequence)

# in update(), while PLAYER_TURN:
if now >= self.turn_deadline:
    self._game_over()
```

5. Files produced

File	Purpose
main.py	Window setup, audio pre-init, main loop
game/game_engine.py	State machine, timed playback, input checking, sound, timer bar, Game Over overlay
game/color_button.py	Pad drawing (dim and lit states) and each pad's pitch
README.md	Lab write-up with run steps, tasks and video links

Full chat link: (paste shared chat URL here)